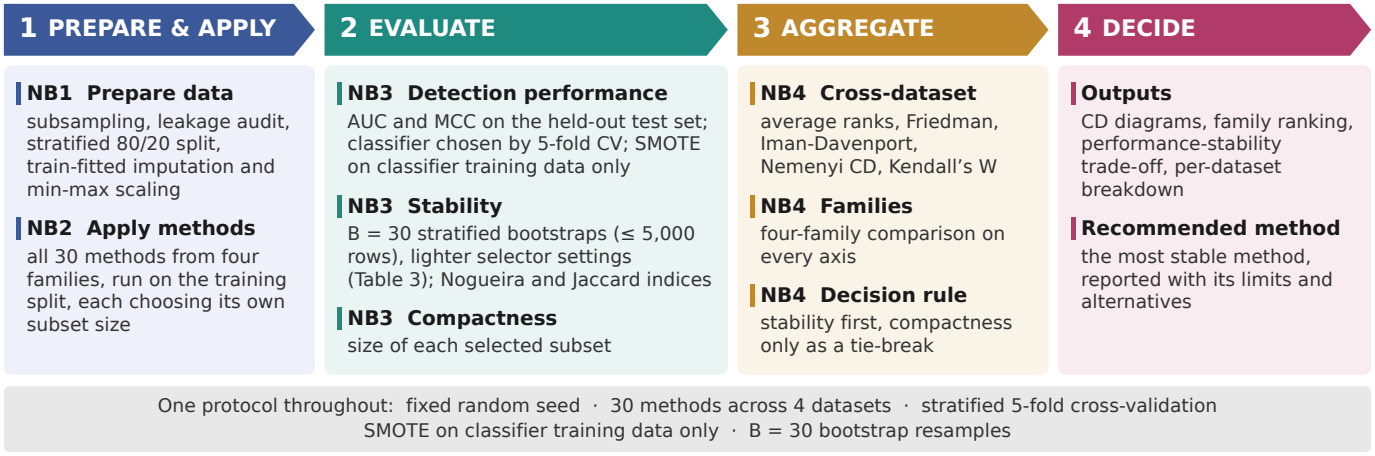
1 PREPARE & APPLY
NB1 Prepare data
subsampling, leakage audit, stratified 80/20 split, train-fitted imputation and min-max scaling
NB2 Apply methods
all 30 methods from four families, run on the training split, each choosing its own subset size
2 EVALUATE
NB3 Detection performance
AUC and MCC on the held-out test set; classifier chosen by 5-fold CV; SMOTE on classifier training data only
NB3 Stability
B = 30 stratified bootstraps (≤ 5,000 rows), lighter selector settings (Table 3); Nogueira and Jaccard indices
NB3 Compactness
size of each selected subset
3 AGGREGATE
NB4 Cross-dataset
average ranks, Friedman, Iman-Davenport, Nemenyi CD, Kendall's W
NB4 Families
four-family comparison on every axis
NB4 Decision rule
stability first, compactness only as a tie-break
4 DECIDE
Outputs
CD diagrams, family ranking, performance-stability trade-off, per-dataset breakdown
Recommended method
the most stable method, reported with its limits and alternatives
One protocol throughout: fixed random seed · 30 methods across 4 datasets · stratified 5-fold cross-validation
SMOTE on classifier training data only · B = 30 bootstrap resamples